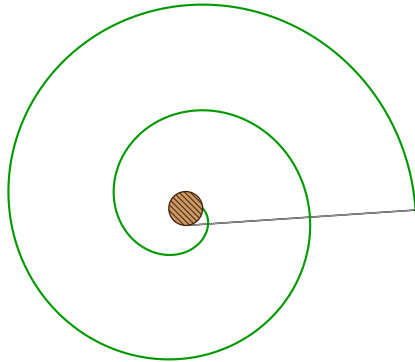
Problem 1 ... mowing

3 point

n an attempt to save effort, Verča decided to automate mowing her awn. She firmly planted a pole in the center of the lawn and attached he lawnmower to it with a rope. The mower can move and cut only orward, but due to the rope winding around the central pole, its path vill form a spiral, allowing it to cut an approximately circular area around he pole. What radius must the pole have so that the rope winds in such way that the cut strips perfectly align with each other? The mower uts strips of grass 0.75 m *wide; neglect the thickness of the rope.*

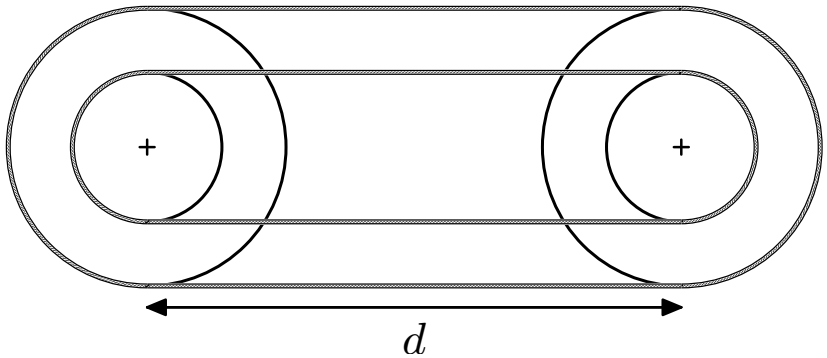


Verča watched DIY videos

Problem 2 ... different period

3 point

Two parallel shafts are spaced $d = 25.0\,\mathrm{cm}$ *apart. Each carries one wheel of radius* $r = 3.50\,\mathrm{cm}$ *and another of radius* $R = 6.50\,\mathrm{cm}$. *A stretched rubber band is looped around the two small wheels, and another around the two large ones. One shaft rotates at frequency* $f = 1.28\,\mathrm{Hz}$. *Find the ratio of the time for the small-wheel band to return to its initial position after one rotation to that for the large-wheel band. The rubber band does not slip on the wheels.*



Jarda was chained by his love for mechanical engineering

Problem 3 ... playing with gas 3 point

By what factor does the volume of an ideal gas change if its thermodynamic temperatur
ncreases by 80 % *and its pressure decreases by* 60 %*?*

Tomáš was gassed after doing so much work

Problem 4 ... fast serve

4 point

In table tennis, it is sometimes tactical to surprise your oponent with a long and fast serve The first impact of the ball is as close as possible to the edge of the serving table, the secon mpact is as close as possible to the opponent's edge. What is the shortest possible ball fligh ime from one side of the table to another? Do not consider air resistance or ball rotation Look up all the necessary parameters.

Jarda wants to serve a defeat as a second course to his opponen

Problem 5 ... old power supply

3 point

David found an old DC power supply. He was wondering about how to use it, so he tried t connect it into a circuit with 2 resistors with the resistance of $R = 10\,\Omega$. In the first case h connected them in series, then parallel. In both cases he measured the current in the circuit n the first case he measured the current as $I_1 = 1.2\,\mathrm{A}$ and in the second one as $I_2 = 4.0\,\mathrm{A}$ David knew that the power supply is not ideal. Calculate the internal resistance of the powe upply.

David sought after some nice electrical problem

Problem 6 ... firmly connected mathematical pendulums

3 point

Consider two massless rods of length $l = 15$ cm, each suspended by one end and able t
otate freely about their suspension points. These suspension points are at the same height
eparated by a distance $d = 30$ cm. The free ends of the rods are connected by a rod o
nass $m = 300$ g and length d. What is the period of small oscillations if the system is displace
within the plane containing the rods? The system is in a gravitational field with acceleration g

Lego idealized a spring as a thin rod

Problem 7 ... icy cold LED lights

3 point

David read a study stating that LEDs in a laboratory have an electrical input power of $P_{\text{ele}} =$ $= 30\,\text{pW}$, *but an optical output power of* $P_{\text{opt}} = 69\,\text{pW}$, *which would correspond to an efficienc* *greater than* 100%. *At first, he was surprised, but then realized that part of the power come* *rom thermal energy. How many diodes would be needed to produce an ice cube with a volum* *f* $V = 10\,\text{cm}^3$ *in one hour, assuming we have water at a temperature of* $20\,^\circ\text{C}$? *Assume tha* *ll optical power comes from the heat of the water and the electrical source. The optical powe* *s independent of temperature, and heat transfer from the surroundings is negligible.*

David did not realize led (Czech for ice) and LED are just homophones

Problem 8 ... lost gas

4 point

In the examination of the surface area of porous or powdery materials, gas adsorption o
he material can be utilized. A sample with a skeletal density of $1200\,\mathrm{kg\cdot m^{-3}}$ *and mass* $59\,\mathrm{m}$
s placed into a $15\,\mathrm{ml}$ *pipette, and the air is evacuated to create a very high vacuum. Then*
$1.1 \cdot 10^{-3}\,\mathrm{mol}$ *of nitrogen gas is introduced After reaching equilibrium, the pressure is mea*
ured to be $0.15\,\mathrm{bar}$*. What is the surface area of the sample, assuming that the area of on*
dsorbed molecule is $0.16\,\mathrm{nm^2}$*? The whole experiment is conducted at the temperature of liqui*
itrogen $T = -196\,^{\circ}\mathrm{C}$*.*

Jarda is now looking for a way to get his gas back

Problem 9 ... brokenly charged 4 point

A particle characterized by a charge $Q = -15.0\,\mu\text{C}$ *and a mass* $M = 50.0\,\mu\text{g}$ *is flying perpen*
licular to magnetic field lines at a speed of $v = 100\,\text{m}\cdot\text{s}^{-1}$. *At one moment, this particle decay*
nto two particles with opposite charges. What is the ratio of the radius of curvature of th
positively charged particle's trajectory to that of the negatively charged particle? The mass o
he resulting positively charged particle is $m = 20.0\,\mu\text{g}$, *and its charge is* $q = 10.0\,\mu\text{C}$. *Assum*
hat after the decay, both particles retain the velocity v, *and they do not interact with eac*
ther further. *Pepa was procrastinating during the history of physics class*

Problem 10 ... creative DnD Hero

4 point

The heroes fight a necromancer in his laboratory. One of the heroes is not exactly strong however, he is creative and so he gets the idea to knock over a wardrobe so that it would collapse onto a nearby undead. In the brink of a moment, he starts wondering whether hi having the strength to knock over the wardrobe implies that the undead would subsequentl have enough strength to hold the wardrobe. Suppose the wardrobe is homogeneous and ha height $a = 3.5\,\mathrm{m}$, width $b = 1.5\,\mathrm{m}$ and depth $c = 60\,\mathrm{cm}$ and the hero can topple it along it width axis. Both the hero and the undead have height $h = 180\,\mathrm{cm}$ (the height at which the will act on the wardrobe). The undead is standing $d = 1.5\,\mathrm{m}$ from the wardrobe (measured from the edge of the wardrobe base that remains on the ground). What is the ratio of the force that the undead needs to hold it over its head (provided it keeps standing in the same place) to the force the hero needs to knock it over? Neglect the non-zero speed of a falling wardrobe i.e., consider this to be a static problem; both the undead and the hero pushed perpendicularly toward the wardrobe side.

Lego has only played DnD a few times

Problem 11 ... staircases in Atomium

3 point

The Atomium in Brusel is a monumental tourist attraction that resembles an elementary ce of a spatially centred lattice of crystalline iron. Instead of atoms, there are large spheres tha an be entered. The spheres are connected by passages running along all edges of the cube an also from the corner spheres into the central sphere. The cube is standing on one of its vertice so that its body diagonal points perpendicular to the ground. What is the slope of the staircas eading from the second lowest sphere to the central sphere with respect to the horizontal plan of the ground?

Jarda was on a holiday in Belgium

Problem 12 ... the strength of LED strips

4 point

Jarda installed 2.5 m *long LED strips on his kitchen counter, with a power consumptio* *of* 11 W/m *and a voltage of* $U = 12\,\mathrm{V}$. *The voltage is supplied from a transformer via a* $l =$ $= 4.8\,\mathrm{m}$ *long twin-lead cable in which the two wires are spaced* $d = 3.0\,\mathrm{mm}$ *apart. What is th* *magnitude of the force between the pair of wires? If the force is attractive, submit a positiv* *value; if repulsive, submit a negative value. Assume the thickness of the metal wires to b* *negligible.*

Jarda was attracted by the electron force in cables while decorating his kitchenette

Problem 13 ... skipping pizza 4 point

On a spring of stiffness $k = 100\,\mathrm{N{\cdot}m^{-1}}$ *with a platform there is a pizza of mass* $m = 0.40\,\mathrm{kg}$. *Th*
pring with the platform is, by a built-in mechanism, compressed to an initial displacement x_0 =
= $0.020\,\mathrm{m}$ *from the equilibrium position. The mechanism then releases the spring, whic*
aunches the pizza and immediately after the launch compresses the spring back to the initia
lisplacement x_0. *The pizza then lands back on the platform, compresses the spring and i*
aunched again. Immediately after each launch the mechanism pulls the spring to the maxima

displacement from the equilibrium position that was reached after the pizza's impact. Thi process repeats.

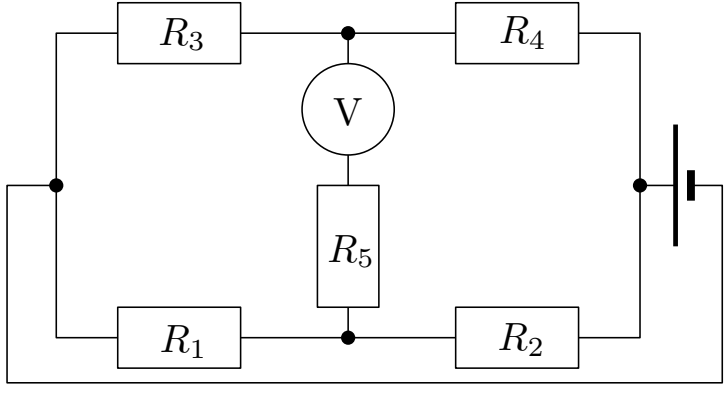
To what displacement will the spring be compressed by the mechanism after the launch a which the pizza first exceeds the height $h = 1.0\,\mathrm{m}$ above the equilibrium position? Neglect ai esistance. Assume that all kinetic energy of the pizza and the potential energy of the sprin re converted into each other ideally (without losses) and that the mass of the platform an he spring is negligible compared to the mass of the pizza.

David was thinking about making the preparation of Hawaiian pizza more efficien

Problem 14 ... voltage on the voltmeter

4 point

What voltage will the voltmeter show in the circuit shown in the diagram? The source voltag
s $U_0 = 4.50\,\mathrm{V}$*, the source has negligible internal resistance, the voltmeter is ideal, and th*
esistors have the following resistance values: $R_1 = 1.00\,\Omega$, $R_2 = 2.00\,\Omega$, $R_3 = 4.00\,\Omega$, R_4 =
$= 2.00\,\Omega$ *and* $R_5 = 5.00\,\Omega$*. Provide the result as a positive number.*



Karel was thinking about using the voltmeter incorrectly

Problem 15 ... radar operator problems

4 point

A radar operator transmitted a pulse at frequency $f = 50\,\mathrm{MHz}$ toward an enemy aircraft flyin
t an altitude $h = 20\,000\,\mathrm{ft}$ above the ground. After a time $t = 0.10\,\mathrm{ms}$ the pulse returned wit
frequency shifted by $\Delta f = 190\,\mathrm{Hz}$ relative to the transmitted pulse. Determine how long i
will take until the enemy aircraft is directly above the radar operator, assuming the aircra
lies uniformly in a straight line and parallel to the ground toward the point directly above th
adar.

Dodo was watching the (weather) radar

Problem 16 ... spinning electrons 4 point

Danka was working in the lab with flask filled with low-pressure argon, into which a collimate
lectron beam was introduced. The beam ionized the argon, and during recombination, th
ath of the electron beam could be observed due to luminescence. Danka adjusted the bea
o that its trajectory took the shape of a helix with a radius of 5.00 cm *and a constant pitc*
f 5.00 cm. *The accelerating voltage of the electrons was* 200 V. *What was the magnitude of th*
agnetic induction (magnetic field) in the flask produced by the external magnets under thes
onditions? Assume that the magnetic induction is constant throughout the entire volume o
he flask and has a uniform direction.

Danka did an experiment for he Practical Course class, but it was a long time ago

Problem 17 ... overturned outhouse

4 point

It was windy and Kuba was walking along a footpath. Suddenly he saw an overturned outhouse and immediately began to wonder what wind speeds are required to tip it over. Approximat he outhouse as a homogeneous rectangular block of height $a = 244\,\mathrm{cm}$*, with a square bas of side length* $b = 152\,\mathrm{cm}$ *and mass* $m = 140\,\mathrm{kg}$*; the drag coefficient has the value* $C = 1.05$ *Assume that the wind blows perpendicularly onto one of the side walls and that the axis o otation is fixed at the edge of the outhouse touching the ground opposite the side on whic he wind is blowing. Take the air density to be* $\rho = 1.225\,\mathrm{kg{\cdot}m^{-3}}$*. What is the minimum win peed required for the outhouse to start tipping?*

Jarda P. went to the relieve himself

Problem 18 ... potassium cube

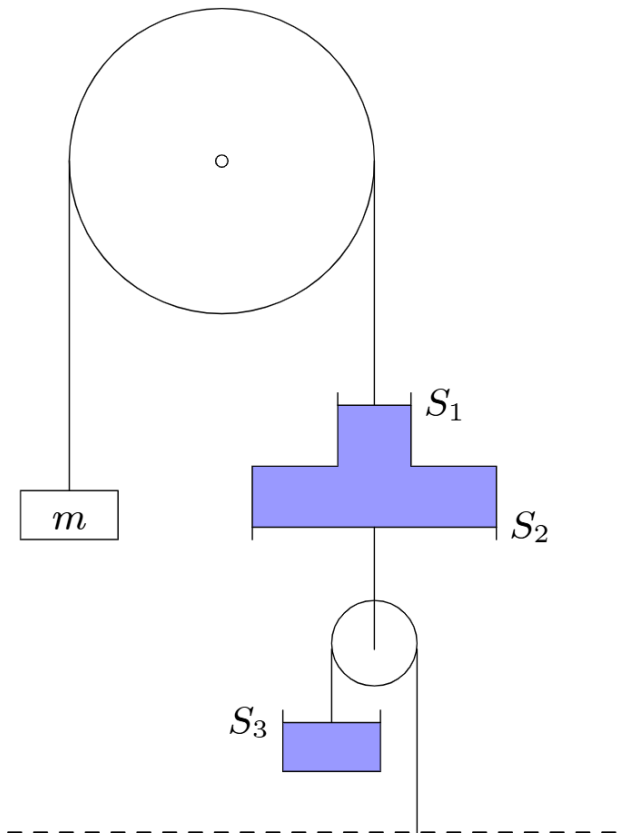
4 point

Potassium $^{40}\mathrm{K}$ *has a half-life of* $T = 1.3 \cdot 10^9$ years. *It transforms via* β^- *decay into stabl* *alcium isotope* $^{40}\mathrm{Ca}$ *with a probability of* 89 %, *and via* β^+ *decay into stable argon isotope* $^{40}\mathrm{A}$ *with a probability of* 11 %. *How much* $^{40}\mathrm{K}$ *would be required in a cube to produce enough argo* *hat would fill a room of dimensions* $a = 5.0\,\mathrm{m}$, $b = 10\,\mathrm{m}$ *and* $c = 3.0\,\mathrm{m}$, *assuming an argo* *particle of* $\rho = 1.0 \cdot 10^{25}\,\mathrm{m}^{-3}$, *in one year? Assume that argon is released from the entir* *olume. The molar mass of* $^{40}\mathrm{K}$ *is* $M = 40\,\mathrm{g{\cdot}mol}^{-1}$.

David wondered how many bananas could he eat before choking on them

Problem 19 ... thermodynamical pulleys 4 point

We have a steady configuration as shown in the figure. Both gas vessels are rigidly fixe and each contains $n = 1\,\mathrm{mol}$ *of gas at a temperature* $T = 20\,^{\circ}\mathrm{C}$*, which is also the ambien temperature where the atmospheric pressure applies. What is the volume of the gas in th lower piston? Neglect the weight of everything except the weight* $m = 2.5\,\mathrm{kg}$*. The areas in th figure are* $S_1 = 15\,\mathrm{cm}^2, S_2 = 50\,\mathrm{cm}^2, S_3 = 25\,\mathrm{cm}^2$*.*

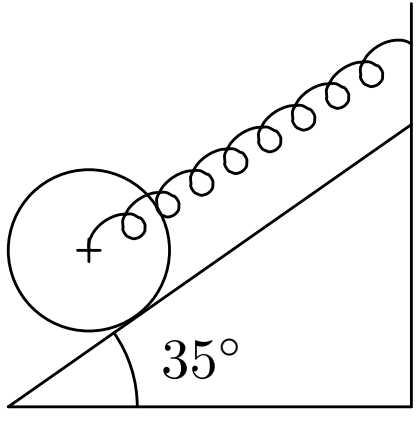


Lego tried to mix-and-match

Problem 20 ... wired wheel

5 point

A hoop with a mass of $1.3\,\mathrm{kg}$ *and a diameter of* $73\,\mathrm{cm}$ *is placed on an inclined plane, tilted a*
n angle of $35\,^\circ$ *to the horizontal. The hoop has an axis of symmetry with a shaft that allow*
t to rotate freely. The shaft is connected to a spring with a spring constant of $2.5\,\mathrm{N{\cdot}m^{-1}}$
which runs parallel to the plane and points upwards. If the hoop is displaced downwards fro
ts equilibrium position, what will be its oscillation period? Assume that there is no slippin
etween the hoop and the plane.



Jarda found a truly unique item in a souvenir shop

Problem 21 ... particles

4 point

We have a relativistic proton for which the detector measured an energy of $E = 5.12 \cdot 10^{-10}$ J
What must be the magnitude of the magnetic flux density B *so that the particle's trajector*
has a radius of curvature $R = 1.11\,\mathrm{m}$*?*

Vítek heard he could possibly avoid taking the Practical Course IV class

Problem 22 ... modified solar system 5 point

Suppose that the Sun had a temperature of $T_2 = 8\,000\,\mathrm{K}$. By what percentage would th orbital period of Jupiter need to increase in order for it to receive the same power output a t currently does? Assume that Jupiter's orbit around the Sun is circular and that all othe properties of the Sun remain unchanged.

Danka has tried to come up with interesting problems

Problem 23 ... harmonical friction

5 point

As is well known, the irregularities of contacting surfaces cause friction. A very simple one dimensional model of such a surface can be represented by the graph of a harmonic functio with a period of $d = 20.0\,\mu\text{m}$ *and an amplitude of* $h = 1.00\,\mu\text{m}$. *If we place two identica "surfaces" of this type against each other, they interlock so that they follow the same curve Determine the coefficient of static friction of such a system—that is, the ratio* F/F_n *between th horizontal force* F *acting on the upper body (at which the system begins to move continuously and the normal force* F_n *pressing the upper body downward. The lower body is fixed and doe not move. The model surfaces are perfectly smooth.*

Kuba tried to formulate a simple friction mode

Problem 24 ... tense oscillations 5 point

Terka was playing with a plate capacitor consisting of two parallel metallic plates separated b a distance $d = 10.0\,\mathrm{cm}$ and held at a potential difference $U = 200\,\mathrm{V}$, when suddenly her earrin fell into the capacitor. She looked in tensely and observed an interesting phenomenon—th earring began to jump back and forth between the plates.

Determine the steady-state frequency of these oscillations. Each plate of the capacitor ha an area $S = 1.00\,\mathrm{m}^2$. Model the earring as a conducting sphere of radius $a = 2.00\,\mathrm{mm}$ an mass $m = 0.15\,\mathrm{g}$. Collisions of the sphere with the plates are characterized by a coefficien

of restitution $\beta = 0.98$. *Upon contact with a plate, the sphere acquires a charge* $q = \pm$
$\pm 4.45 \cdot 10^{-11}$ C. *Neglect the influence of gravity on the motion.*

David misses doing experiments in a lab

Problem 25 ... bizzare yo-yo adventure 5 point

A yo-yo is made of two cylinders with radius R connected by a small axle with radius r. The ratio between R and r is $q = 3.00$. All three cylinders have the same width; they are homogeneous, made of the same material. A massless, infinitely thin string of length $L = 1.25\,\mathrm{m}$ is wound around the small axle, with one end of the string firmly attached to it. When the yo-yo is released and allowed to unwind freely, after what time τ will it return to its original position? Assume that at the moment of release, the tangent to the free end of the string is vertical. Also assume that $r \ll L$, so that the effects at the turning point of the yo-yo's motion can be neglected. *Petr was brainstroming competition problems*

Problem 26 ... falling square 5 point

A very long, vertically oriented rod of negligible diameter carries a current $I = 1.7\,\mathrm{A}$ *upward*
A thin conductive square of side $a = 15\,\mathrm{cm}$*, linear resistance* $\lambda = 230\,\mathrm{m\Omega \cdot m^{-1}}$*, and mass* $12\,\mathrm{g}$ *i*
placed symmetrically around the rod and released. Determine the speed of the square after $7.5\,\mathrm{s}$

Jarda is trying to stop his things from falling

Problem 27 ... ion thrust engine

4 point

There is currently work on the concept of a space drive that accelerates a rocket by ejectin
plasma accelerated to relativistic speeds. Consider a rocket with a total mass of 100 t *tha*
ve accelerate by ejecting protons. Each proton has a total energy of 1 TeV *and* 10^{-3} mol *o*
protons are emitted from the rocket per second. How long will it take to accelerate the rocke
o a speed of 100 km/s *in this way? Assume the rocket's mass remains approximately constan*
during the acceleration.

Jirka listened to a very interesting podcas

Problem 28 ... action scene with motorbike and truck

5 point

A block with a mass of $M = 123\,\mathrm{kg}$ *moves without friction at a constant velocity of* v = $= 7.35\,\mathrm{m{\cdot}s^{-1}}$ *along a horizontal surface. On top of the block is a cylinder with a mass of* m = $= 11.0\,\mathrm{kg}$, *which rolls with a velocity of* $u = 2.23\,\mathrm{m{\cdot}s^{-1}}$ *relative to the block in the direction o he block's motion. The cylinder, therefore, eventually rolls off the block and begins to mov long the plane as well, assuming that the vertical component of its velocity is absorbed upo mpact. How long after hitting the surface will the cylinder collide with the block? Assum hat their mutual distance at the moment of impact was negligible. The coefficient of kineti riction between the cylinder and the plane is* $f = 0.278$.

Vašek is already starting to come up with physics problems

Problem 29 ... water turn 6 point

Consider a boat of mass $m = 900\,\mathrm{kg}$ *shaped as a rectangular prism with length* $a = 4.0\,\mathrm{m}$ *width* $b = 1.2\,\mathrm{m}$, *and height* $c = 80\,\mathrm{cm}$. *The boat is propelled by a propeller that deliver an angular momentum of magnitude* $L = 400\,\mathrm{J{\cdot}s}$ *directed toward the bow. On the wate surface, the boat turns with an angular velocity of* $10\,^\circ{\cdot}\mathrm{s}^{-1}$ *to the right. What is the angl between the water surface and the deck during this maneuver? If the bow is higher than th tern, give a positive value. Assume the center of mass of the boat is at its geometric cente*

Jarda would like to ride a flying came

Problem 30 ... chasing the Sun

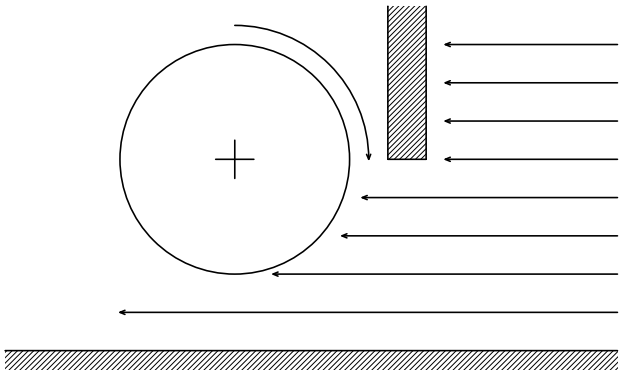
5 point

At 6:10 AM, Danka is sitting in an airplane taking off from Prague Airport, heading directl
ast. The airplane ascends at an angle of $30\,^\circ$ *with a speed of* $300\,\mathrm{km\cdot h^{-1}}$*. On this day, th*
unrise in Prague is expected to occur at 6:38 AM, but Danka will witness it earlier from th
lane. How many minutes after takeoff will Danka observe the sunrise? For the purposes o
his problem, Prague has been moved to the equator, and Danka is flying on the equinox. Also
onsider the Earth to be a perfect sphere, with a 24-hour day, and neglect the Earth's motio
round the Sun.

Thanks to flying, Danka has experienced an earlier sunrise several times

Problem 31 ... gas flow

5 point



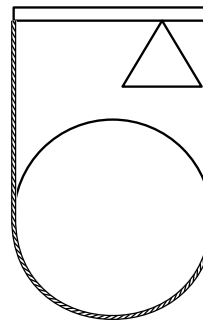
Consider a cylinder of radius $r = 3.00\,\mathrm{mm}$ *and height* $h =$ $= 5.00\,\mathrm{mm}$, *which can rotate about its axis of symmetry. Gas is directed at the cylinder so that particles strike only one half of he cylinder. The other half is covered (the boundary between he halves is the plane that contains the cylinder's symmetry xis, see the figure). The cylinder spins up to a frequency* $f =$ $= 65.0\,\mathrm{Hz}$. *Assume that nothing strikes the cylinder from the other side. A resistive torque* $M = 2.00 \cdot 10^{-9}\,\mathrm{N{\cdot}m}$ *acts on its rotation such that upon impact gas molecule changes its velocity to the velocity of the cylinder surface at that point. The ga lows uniformly across the cross-section that hits the cylinder. What amount of substance of ga in mol per second) strikes the cylinder? The gas is argon with a volumetric concentration* $c =$ $= 2.00 \cdot 10^{-4}\,\mathrm{mol \cdot m^{-3}}$.

Jarda likes to direct gas streams at objects during his free time

Problem 32 ... warming up by work 5 point

We have a piston filled with air (i.e., the Poisson's constant is equal to $\kappa = 1,4$) unde tandard conditions. The surroundings are also at standard conditions, and the piston wall re thermally conductive. We want to bring the air to a state where it has twice the temperatur in the thermodynamic sense) and the same volume as it has now. We have a heater that ca upply the necessary heat; however, we want to conserve the heater and its energy for toug imes. Therefore, we aim to partially supply the energy to the air through mechanical work Specifically, we can expand the piston volume to twice its current volume (and then compres t back), and we can use the heater at any point during this process. What is the minimun mount of energy we must supply via the heater? Express your answer as a percentage of th mount we would need to provide if we did not perform any mechanical work and only use heater. *Lego remembered inexact differentails used in thermodynamics*

Problem 33 ... lever–pulleys 5 point



We have a pulley and a lever connected by a massless rope as shown in the figure. Treat the lever as a rod of length $2r = 20\,\mathrm{cm}$ *with mass* $m = 300\,\mathrm{g}$*. The lever is supported so that the lengths of its arms are in the ratio* $3 : 1$*. The pulley has the shape of a disk of radius* $r = 10\,\mathrm{cm}$ *and mass* $M = 600\,\mathrm{g}$*. Both the lever and the pulley are homogeneous and the rope does not slip on the pulley. What is the acceleration of the center of mass of the pulley at the moment of release from the state shown in the figure?*

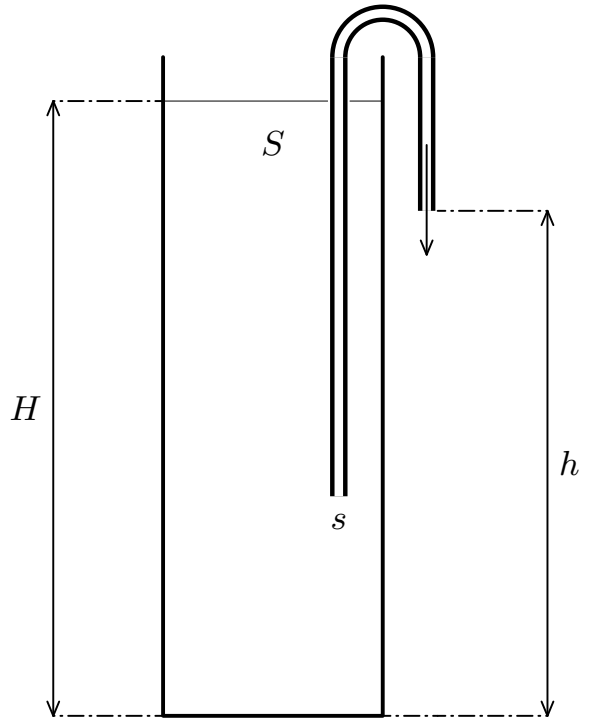
Lego has never tried to combine these simple machines together

Problem 34 ... a straw

6 point

Jarda was playing with his cocktail. He took a straw and fully submerged it in the cocktail. Then he pulled out one end, and to keep the liquid inside, he held the pulled-out end with his finger so that the cocktail would not spill out. Then he released it, and the cocktail began to spill onto the table, much to the displeasure of everyone sitting around.

Calculate how long this calamity will last if Jarda does not intervene. We know that the horizontal cross-section of the glass is $S = 16\,\mathrm{cm}^2$ *and the straw has a constant cross-section* $s = 0.40\,\mathrm{cm}^2$ *along its entire length. Initially, the liquid level in the glass is at a height* $H = 8.0\,\mathrm{cm}$ *from the bottom, and the straw's outlet is outside the glass at a height* $h = 5.0\,\mathrm{cm}$ *above the bottom of the glass.*



Assume that the cocktail is an ideal incompressible fluid. Neglect any undesirable effects such as bubble intrusion or surface tension. The straw does not move while the liquid flows out.

Jarda has a remarkable ability of spilling anything he tries to drink

Problem 35 ... waging a war with windmills

6 point

Don Quixote calmly rides on his faithful Rocinante through Mancha, when suddenly he spot a four-armed giant on the horizon, waving threateningly at him with all four limbs. Howeve we, who have our wits about us, clearly see that it is not a giant, but a windmill. It has fou blades with a rectangular cross-sectional area $S = 300\,\mathrm{cm}^2$ *and surface area* $P = 24\,\mathrm{m}^2$*, whic are placed on rigid rods at a distance* $d = 1.00\,\mathrm{m}$ *from the axis of rotation of the mill an have length* $L = 6.00\,\mathrm{m}$*. They are made of homogeneous material with Young's modulus* $E =$ $= 1.00\,\mathrm{MPa}$ *and each has mass* $M = 170\,\mathrm{kg}$*. The "giant's threatening" waving was caused b a gust of wind, which spun the blades to an angular velocity* $\omega = 1.60\,\mathrm{s}^{-1}$*. To what length L do the blades elongate? Assume that these are small deformations and they occur only in th radial direction; also assume that the width and thickness of the blades are small compare to the distance from the axis and thus that the magnitude of the centrifugal force depend only on the perpendicular distance from the end of the blade. Do not consider the influence o gravitational force on the deformation of the blades.*

Petr was reminiscing about the Mechanics course

Problem 36 ... conducting corner

7 point

The boundary of the first octant of the Cartesian coordinate system forms a conductin corner (i.e., it consists of three quarter-planes, each oriented along one of the x, y, and directions). A charge q moves along the half-line $x = y = z > 0$. Determine the ratio of th charge Q placed fixed at the origin to the charge q such that the conducting corner can b replaced by Q while the force acting on q remains unchanged. You may neglect the induce magnetic field.

Kuba was sent to the corner as a punishmen

Problem 37 ... FTL

6 point

Consider an empty container formed by rotating the function $y = x^{-3}$ around the axis $x =$ for positive values of y, with all dimensions in meters. The container is closed at the botton by the rotated x-axis. Water flows into the container at a rate of $Q = 200\,\mathrm{l{\cdot}s^{-1}}$. Assumin the water instantly forms a flat surface, after how long would the water level begin to rise a superluminal speed?

Marek wanted to be a buoy

Problem 38 ... speed of a shadow

6 point

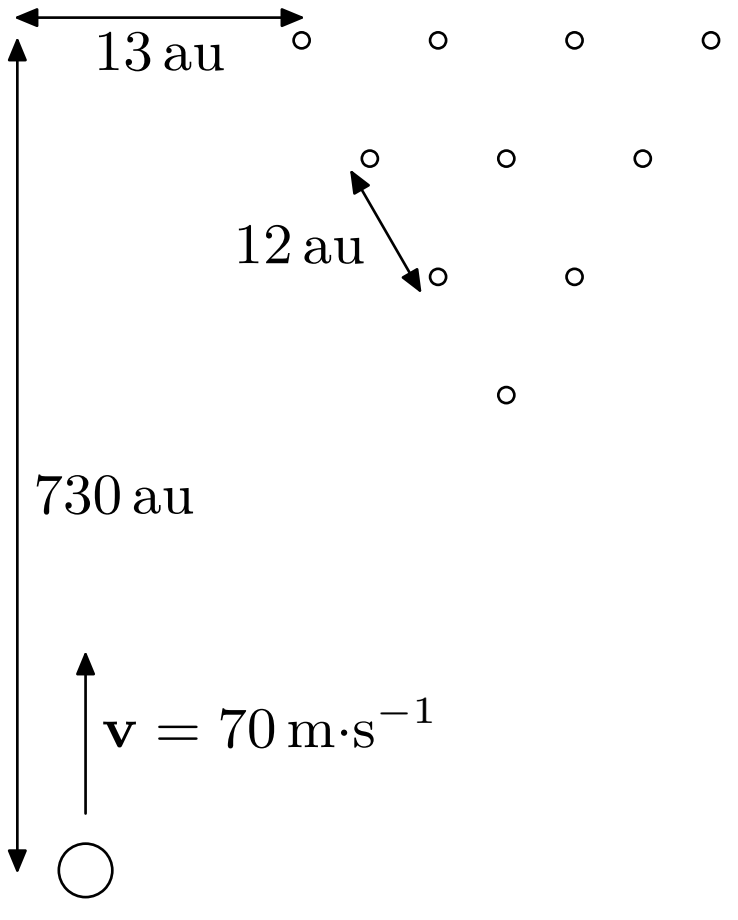
Jarda played tennis in the sun, which was $32\,^\circ$ *above the horizon. He served the ball to hi*
pponent by hitting it from a height of 70 cm *at an angle of* $38\,^\circ$ *to the ground. The ball landed*
n the other side of the court at a distance of 25 m *from Jarda. Determine the maximum speed*
f the shadow that the ball casts on the ground. The plane of the ball's trajectory is tilted at
n angle of $23\,^\circ$ *with respect to the vertical plane in which the sun's rays strike the ball.*

When Jarda's forehand stopped working, he unleashed physics on his opponent

Problem 39 ... heavenly bowling

7 point

t has been a while since God created the world, and be-ause sometimes He has nothing to do, he likes to play owling with planets. On the Milky Way, he places ten mall planets in a classical equilateral triangle, so that the elative distance between two planets on the edge is 12 au. *For the ball, God choses a star the size of our sun, which he hrows in the plane of the "pins" from a distance of* 730 au *rom the back row of planets toward them. Although he gave his sphere the necessary rotation, he forgot that there s no friction in space. The star thus did not turn toward he planets, but instead flew parallel to the axis of the riginal triangle at a distance of* 13 au *from one of its cor-ners at a speed of* $70\,\mathrm{m{\cdot}s^{-1}}$. *Nevertheless, the star did trike some of the planets. How many planets did God "knock off" in this way? The planets are so small that heir gravitational interaction is negligible. Also, do not onsider the effect they have on the star's orbit.*



Jarda would like to throw like a goo

Problem 40 ... leaking liquids

6 point

Consider a cylindrical container with the cross-section of $S = 42.0\,\mathrm{cm}^2$. *Water is poure*
nto the container to a height of $h = 5.0\,\mathrm{cm}$ *above which is olive oil with a density of* $\rho_{\mathrm{oil}} =$
$= 910\,\mathrm{kg{\cdot}m}^{-3}$ *with the same layer thickness* h, *and ethanol above the oil with, also with th*
ayer thickness h *and a density of* $\rho_{\mathrm{ethanol}} = 789\,\mathrm{kg{\cdot}m}^{-3}$. *If a hole with an area of* $s = 4.2\,\mathrm{mm}^2$ *i*
pened at the bottom of the container, how long will it take for the water and oil to drain
eaving only the ethanol in the container?

Neglect surface tension and cohesion of the liquids. A numerical simulation is recommended

Karel varied and complicated the problems

Problem 41 ... the cosmos getting far, far away

7 point

How far away would stars and galaxies have to be – objects that would otherwise be relativel
t rest with respect to us, but which we observe with such a redshift due to the expansion o
he universe that the wavelengths of their radiation are twice as large as at the moment the
vere emitted? We are asking about the "present-day" distance, not the distance they had a
he time the signal was emitted. Assume that the universe is homogeneous and isotropic, tha
Einstein's equations and the standard ΛCDM cosmological model hold, and that the univers
xpands with a Hubble constant that is constant in time, with value $H_0 = 68\,\mathrm{km{\cdot}s^{-1}{\cdot}Mpc^{-1}}$.

Karel was thinking about a very distant zone

Problem 42 ... underwater light

7 point

Consider a point source of unpolarized light radiating in all directions. The source is submerge
at a depth $h = 1.00\,\mathrm{m}$ below a (sufficiently large) calm water surface; the bottom beneath it i
perfectly dark. What percentage of the radiant power is transmitted above the surface? Th
refractive index of water is $n_i = 1,33$; the refractive index of air is $n_t = 1.00$. Assume that th
water absorbs none of the light. Approximate the intensity transmission coefficient by its valu
at normal incidence everywhere where light is transmitted.

Petr was reminiscing about the Optics course

Problem 43 ... hmm, where did i put that? 8 point

You've surely noticed the increase in entropy in your room, or that things tend to diffus especially when your roommate is away). Recently, Lego stored his favorite Brownian particl in the minimum of a 2D potential shaped as $U(x, y) = C(x^2+y^2)$*, where* $C = 1.0 \cdot 10^{-20}\,\mathrm{J \cdot m^{-2}}$ *After a very long time, he went to look for it. What is the probability that he finds it withi a distance less than* $d = 50\,\mathrm{cm}$ *from the spot where he left it? This task was conceived durin the summer, so the temperature in the dorm room is* $T = 40\,^\circ\mathrm{C}$*.*

As a theoretical physicist, Lego loves physics that have something to do with reality

Problem 44 ... racing against the rain

6 point

David and Terka enjoy playing catch with a ball. During the summer, they were able t throw it a maximum distance of $5.0\,\mathrm{m}$. *The ball has a radius of* $r = 15\,\mathrm{cm}$ *and a mass of* $m =$ $= 300\,\mathrm{g}$. *By how many centimeters (in absolute value) will the distance they can throw th ball differ in winter? The air density outside the ball is in summer* $\rho_1 = 1.12\,\mathrm{kg}\cdot\mathrm{m}^{-3}$ *and* $\rho_2 =$ $= 1.29\,\mathrm{kg}\cdot\mathrm{m}^{-3}$ *in winter. The atmosphere does not affect the athletic abilities of David o Terka, nor the physical properties of the ball.*

David was thinking who will play catch with him when it rains

Problem 45 ... blowing up a balloon reloaded

7 point

When inflating a balloon, we can, from a certain radius onward, assume with good accurac hat the elastic energy increases directly proportionally to the growing surface area of th balloon. In this dimension range, Jirka found his previously inflated balloon, which in th meantime had deflated to a radius of 5.0 cm *(the balloon is approximately spherical). Sinc he balloon is rather small, he decided to inflate it back to its original size. Balloon inflatio s, however, physically demanding. Jirka wants to know how much work his muscles perfor when he inflates the balloon using a maximal possible exhalation.*

Assume that after inhaling, Jirka has 5.5 l *of air in his lungs, and after maximal exhalation .5 l of air remain in his lungs. Further assume that the pressure and temperature of the ai n his lungs is at all times equal to those in the balloon, and approximate the inflation as a sothermal process (thus we consider the lungs and the balloon as a single connected vesse undergoing a quasistatic process). The air temperature in the balloon is* 25 °C *and the initia pressure* 107 kPa*. This pressure value corresponds both to the situation before the inflatio begins (i.e., for the isolated balloon) and at its start. The pressure in Jirka's lungs adjusts t match the initial pressure in the balloon. Compute the work of Jirka's lungs only from th moment the inflation starts. Assume that the work is determined solely by the change in lun volume and the difference between the gas pressure and atmospheric pressure (the surroundin tmospheric pressure assists the lungs during gas compression).*

Jirka ressurected a problem from two years ago

Problem 46 ... a spring and a magnet

8 point

Consider a spring of stiffness $k = 50.0\,\mathrm{mN{\cdot}cm^{-1}}$*, hanging vertically, on which a magnet o*
mass $M = 110\,\mathrm{g}$ *and magnetic moment* $m_1 = 2.00\,\mathrm{A{\cdot}m^2}$ *is attached. The magnet also acts a*
weight. On the floor below the magnetic weight is placed another magnet with the magneti
moment $m_2 = 10.0\,\mathrm{A{\cdot}m^2}$*. Calculate the period of the small oscillations, assuming that th*
magnets are of negligible size, and thus can be considered as magnetic dipoles, and are movin
lowly enough that we can neglect any losses. The magnets are oriented in such a direction tha
hey repel each other, and the magnet acting as a weight is exactly such that the deflection o
he spring does not change when it is suspended.

Magnets are attracted to each other and their relationship can even be harmonic

Problem 47 ... underwater light reloaded

9 point

Consider a point source of unpolarized light radiating in all directions. The source is submerge
t a depth of $h = 1.00\,\mathrm{m}$ *below a (sufficiently large) calm water surface. What percentage o*
he radiant power is transmitted above the surface? The refractive index of water is $n_i = 1.33$
he refractive index of air is $n_t = 1.00$*. Assume that water absorbs none of the light and us*
he exact values of the transmission coefficients.

Petr liked the graph

Problem 48 ... thermodynamic lottery

8 point

Two Maxwell demons, Elenka and Tomáš, work in corporation called "Ensembles Ltd.". Thei ask is to find certain microstates. They both have an isolated system in equilibrium compose of a box with $N = 1.00 \cdot 10^7$ monatomic ideal gas particles and volume V.

Now, each of them looks into their box and checks if the system is in the desired rang of microstates. If yes, these boxes are quickly shipped to the customers. The employee wit higher number of shipped boxes gets a promotion. Elenka really wants to be promoted, so sh heated up Tomáš's box just a little (she is a demon after all).

Consider that the desired range of microstates is the same for both of them and tha each microstate compatible with the macroscopic quantities is equally probable. Furthermore

Elenka's box is at temperature $T_\mathrm{e} = T$ and Tomáš's is at $T_\mathrm{t} = 1.000\,01T$, what is the rati
between the probability that Elenka looks and finds a desired microstate and the probabilit
hat Tomáš does? Consider a classical setting (T and N are large enough).

Marek J. wants to be a Maxwell's demon

Problem 49 ... giant metronome

9 point

Consider a metronome consisting of two rods. The longer rod is homogeneous, with a length of $L = 25.0\,\mathrm{m}$ *and a linear density of* $\lambda = 100\,\mathrm{kg{\cdot}m^{-1}}$*, and is fixed at one end at a point around which it can freely rotate. The shorter rod, with a length of* $l = 10.0\,\mathrm{m}$*, is massless and its upper end is attached to the midpoint of the longer rod, such that it can freely rotate around the point of connection. Moreover, the shorter rod always passes through a point located at a height of* $h = 11.0\,\mathrm{m}$ *above the point where the lower end of the longer rod is fixed – at this point, it can both rotate and slide. At the other end of the shorter rod, there is a weight with a mass of* $M = 2.00\,\mathrm{t}$*. What is the period of small oscillations?*

Lego was at the Prague metronome

Problem 50 ... charged hoop in a sphere

9 point

Consider a hollow, ideally conducting sphere of radius $R = 9.0\,\mathrm{m}$. *A point charge* $Q = 6.0\,\mathrm{n}$
s placed at its center. Next, a circular ring of radius $\rho = 7.0\,\mathrm{m}$ *is placed inside the sphere s*
hat its axis passes through the center of the sphere and the distance between the center of th
phere and the center of the ring is $r = 5.0\,\mathrm{m}$. *The ring is uniformly charged with linear charg*
lensity $\lambda = 3.0\,\mu\mathrm{C}\cdot\mathrm{m}^{-1}$. *A test charge with charge-to-mass ratio* $q/m = 0.30\,\mathrm{C}\cdot\mathrm{kg}^{-1}$ *(assume*
ot to affect the charge distribution on the sphere) is placed on the line segment connectin
he centers of the sphere and the ring. The test charge is allowed to stabilize at its equilibriu
osition inside the sphere. If the test charge is then slightly displaced, what will be the period T
f its small oscillations? *Petr tried to upcycle his Fyziklani problem from last yea*

Problem 51 ... Easterly

8 point

Kuba was successful on Easter Monday; because of that, a heavy cluster of ribbons was locate
t the end of his Easter whip. Determine the weight of this cluster, assuming the willow whi
s a perfectly thin, massless, homogeneous rod of length $l = 1.0\,\mathrm{m}$ *(with ends* A *and* B*) an*
ending stiffness $K = 0.80\,\mathrm{N{\cdot}m^2}$*; when the whip was held vertically at the end* A *(with th*
ibbons tied to the end B*), it was deflected from the vertical axis by an angle* $\theta_0 = 45\,^\circ$*, an*
he tangent at the free end B *of the whip made an angle* θ_0 *with the vertical, while the tangen*
t the held end A *was vertical. The bending stiffness is defined by* $\tau = K\ \kappa$*, where* τ *is th*
xternal bending moment and κ *is the rod curvature.*

Kuba was inspired by the traditional mistreatment of women

Problem 52 ... thermal neutrons

9 point

Jindra wants to detect thermal neutrons more than anything else in the world. For this reason he acquired a semiconductor pixel particle detector and lithium fluoride LiF. Jindra's detecto is a semiconductor slab with a layer of lithium fluoride on its top surface. Thermal neutron hit the layer of lithium fluoride perpendicularly.

The lithium isotope ^{6}Li reacts with thermal neutrons to produce an alpha particle an a tritium nucleus

$$^6\mathrm{Li} + \mathrm{n} \longrightarrow {}^4\mathrm{He} + {}^3\mathrm{H}.$$

The cross-section of the reaction is $\sigma = 940\,\mathrm{b}$. The tritium nucleus is emitted with a kineti energy of $2.73\,\mathrm{MeV}$, and the alpha particle is emitted with a kinetic energy of $2.05\,\mathrm{MeV}$. Jindra'

letector indirectly detects neutrons by detecting alpha particles and tritium nuclei. The numbe of direct interactions of neutrons in the semiconductor is negligible. You may assume that ever alpha particle or tritium nucleus entering the semiconductor detector is detected. The rang of an alpha particle with energy $2.05\,\mathrm{MeV}$ *in lithium fluoride is* $l_\alpha = 5.94\,\mu\mathrm{m}$. *The range o tritium nucleus with energy* $2.73\,\mathrm{MeV}$ *in lithium fluoride is* $l_\mathrm{H} = 32.9\,\mu\mathrm{m}$. *The particle path are straight and do not bend.*

Jindra wants to find such a thickness of the LiF layer that he achieves the highest possibl neutron detection efficiency (i.e., the number of detected neutrons divided by the number o ncident neutrons), yet at the same time, he does not want to use more LiF than necessary What is the highest achievable detection efficiency?

The density of lithium fluoride is $\rho = 2\,635\,\mathrm{kg{\cdot}m^{-3}}$. *Lithium contains* $p = 7.50\,\%$ *of th sotope* ^6Li *atoms. The rest are atoms of the isotope* ^7Li, *which do not react with neutrons.*

Jindr

as not yet acquired a thermal-neutron detector, ho only modelelled the problem in Geant4

Problem 53 ... attractive Petr

8 point

As we well know, David repels women with a force that acts radially outward from him an depends only on the distance with a constant of proportionality of $120\,\mathrm{N{\cdot}m^2}$*. On the othe hand, there is always a cluster of women around Petr. David wanted to measure how Pet ttracts women and found that Petr's charm is a radial attractive force with a constant o proportionality of* $30.0\,\mathrm{N{\cdot}m^3}$*.*

Now, David is standing in a plane surface at a distance of $1.00\,\mathrm{m}$ *from Peter, when a woma ppears in the same plane near them. Determine the area of the region in which she can appea o that she would be attracted to Peter, i.e., so that the resulting force acting on her woul orm an angle smaller than a right angle with the line connecting her and Petr. Do not conside ny other forces than those mentioned.*

David was thinking about how to follow up on a problem from the Physics Braw

Problem 54 ... the fullest spoon

9 point

The inner volume of a spoon is an ellipsoid with semi-axes $a = 4.0\,\mathrm{cm}$, $b = 3.0\,\mathrm{cm}$ *and* $c =$ $= (2.0 + \sqrt{3})\,\mathrm{cm}$ *and height* $h = 5.0\,\mathrm{mm}$ *(see figure). Let us pour salt on it, which has a critica ngle of repose* $\theta = 40°$*. What is the maximum volume of salt we can carry on the spoon?*

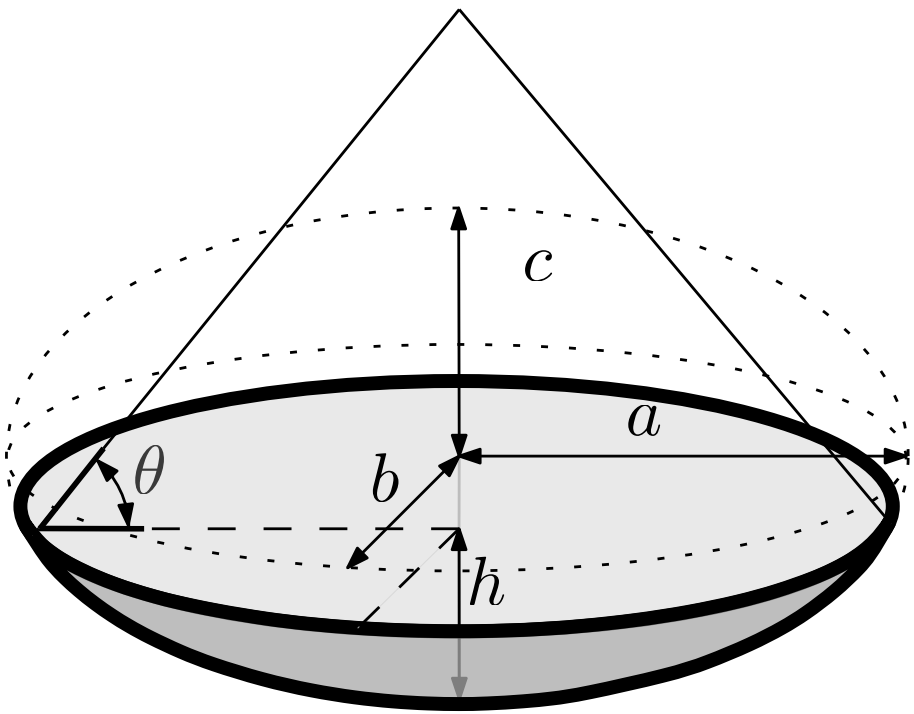


Figure 9: Schema of the spoon.

Jarda always puts just one sugar into his tea

Problem 55 ... Faraday's collector

8 point

Consider a conductive circular target of radius $r = 5.00\,\mathrm{mm}$, *onto which positively charged ion* *mpinge perpendicularly with areal flux* $j = 2.00 \cdot 10^8\,\mathrm{cm}^{-2}\cdot\mathrm{s}^{-1}$ *and charge magnitude equa* *o the elementary charge. However, upon impact of one ion, on average* $\sigma = 0.230$ *secondar* *lectrons are released, which leave the surface with a distribution described by the probabilit* *lensity* $f = (1/\pi)\sin\alpha$ *per unit solid angle with respect to spherical coordinates centered at th* *oint of impact. The angle* α *here is measured from the surface of the disk toward the norma* *What current do we measure in the grounded linear conductor connected to the disk if we als* *make a fence around the target of height* $h = 2.00\,\mathrm{mm}$, *which is conductively connected to th* *lisk and is able to capture these electrons? Assume that the motions of the particles are no* *nfluenced by any electric field.*

Problem 56 ... escape across a frozen dam

9 point

A hiker was trying to flee from a hungry wolf in winter, and his path led past a lowered froze dam. The animal was at his heels, so he had to plan his route as efficiently as possible. He wa going to climb the concrete dam, where he would be safe. There were thick reeds all along th bank, but suddenly, he saw a clear area of ice. He ran onto the frozen surface at $u = 6.00\,\mathrm{m{\cdot}s^{-}}$ *n a direction parallel to the dam, from which he was* $d = 150\,\mathrm{m}$ *away. What is the shortes ime to reach safety if the coefficient of friction between the ice and his boots is* $f = 0.220$ *an f he has to stop at the dam to climb it, which will take another* $10.0\,\mathrm{s}$*?*

Jarda participated in the Physics Cup

Problem B.1 ... mighty power bank in action

3 point

We are buying a 30 000 mA·h *power bank for* 30 USD. *What is the minimum fraction of th otal price represented by the lithium it contains, given that the market price of lithium i* $1.4\,\mathrm{USD\cdot kg^{-1}}$*? The power bank operates using Li-ion battery technology.*

Jarda was frightened by the increase in the price of lithium

Problem B.2 ... heavy charging

3 point

Manufacturers may consider lithium still quite expensive, and so they begin producing powe
banks that use sodium. By how much would the mass of a power bank increase to have th
ame capacity of $30\,000\,\mathrm{mA{\cdot}h}$*? All other components of the device remain unchanged.*

Jarda wondered whether his weary back would wear ou

Problem B.3 ... I don't have the capacity for that

4 point

A $30\,000\,\mathrm{mA{\cdot}h}$ *power bank is connected to a DC source of* $5.0\,\mathrm{V}$ *through a* $1.5\,\mathrm{m}$ *cable. I harges from* 0 *to* 50% *capacity in* $6.5\,\mathrm{h}$. *Determine the cable's resistance per unit length (mea ured in* $\Omega{\cdot}\mathrm{m}^{-1}$*), assuming that the charging circuitry of the power bank requires a constan oltage of* $4.1\,\mathrm{V}$, *independent of the current.*

Jarda's batteries ran out after a demanding week

Problem B.4 ... (in)efficient powerbank

4 point

The electrochemical reactions in our 30 000 mA·h *power bank provide a voltage of* 3.80 V
However, the output generates a USB voltage of 5.00 V. *If the energy loss during this conversio*
s 15.0%, *what is the effective capacity of the power bank for charging a mobile phone?*

Jarda did not recharge

Problem P.1 ... knocking the opposing boule away

4 point

Jirka is playing pétanque with Jarda. Unfortunately, Jarda is currently winning, meaning tha one of his boules is very close to the small wooden target ball (the so-called jack). The onl way Jirka can turn the game in his favor is to knock Jarda's boule away so that it ends u as far from the jack as possible. Therefore, he throws his boule so that it hits Jarda's boul with a horizontal speed of $40\,\mathrm{cm\cdot s^{-1}}$*. How far will Jarda's boule roll if, after the collision, i is acted upon by a constant resistive force of magnitude* $F = \alpha mg$*, where* $m = 700\,\mathrm{g}$ *is th mass of the boule,* g *is the gravitational acceleration, and* $\alpha = 0.010$*? Both boules are identica The collision is perfectly elastic, and after the collision the boules move along the same straigh ine as Jirka's boule before impact (so the collision can be treated as a one-dimensional elasti ollision of point masses).*

Jirka never played pétanque with Jirka

Problem P.2 ... slowing our ball down

5 point

After Jirka's brilliant throw in the ongoing pétanque match, Jarda lost his advantage. Howeve
t is now his turn, and he is devising a risky plan. One of Jirka's boules is located just 5.0 cm
rom the jack. Jarda knows that the collision of the boules is not perfectly elastic, so his boul
vill continue moving at a low speed even after a collision. Jarda intends to throw his boule i
uch a way that it will deflect Jirka's boule and still roll to the jack (Jarda believes that Jirka'
leflected boule will not knock the jack away). What must be the speed of Jarda's boule jus
efore the collision in order for it to travel the distance of 5.0 cm *after the collision? Conside*
he collision as a collision of identical point masses in a straight line. A resistive force acts o
he rolling boule with magnitude $F = \alpha mg$*, where* $\alpha = 0.010$*, as in the previous problem. Jard*

measured before the game that two identical pétanque boules always lose kinetic energy ΔE =
= $0.25\,\Delta E_{\max}$ during a straight-line collision, where $\Delta E_{\max}$ is the maximum amount of kineti
nergy that the identical boules could lose in a collision without violating the law of conservatio
f momentum. If Jirka were to play pétanque against Jirka, he would surely wi

Problem P.3 ... scattered balls

3 point

Tension between Jirka and Jarda is rising. In their ongoing game of pétanque, there are onl
a few throws left, and Jarda has taken the lead thanks to his trick with a partially elasti
collision. For Jirka to make up the deficit, he must likewise exploit more complex physica
behavior in collisions of the boules. He therefore throws his next boule so that it does no
collide with Jarda's boule along a straight line, but "in a plane". What is the maximum angl
hat the lines of the boules' trajectories can form after the collision? In other words, what i
he maximum angle between the vector lines of the boules' velocities after the collision, if w
model it as a collision of identical point masses in the plane, assuming only conservation o
nergy and momentum? Only consider situations in which both boules have nonzero velocitie
fter the collision. Jirka promised that the solution would be much nicer than its statemen

Problem P.4 ... scattered balls reloaded

5 point

Jirka planned his last throw in the ongoing game of pétanque so that after the collision, hi and Jarda's boules would roll apart such that both trajectories form an angle of $45\,^\circ$ *with th original direction of Jirka's boule (in the previous problem you convinced yourselves that thi s indeed possible for the collision of two point masses in a plane). However, in his planning Jirka forgot that pétanque boules are not point masses but rigid bodies, which, in the case o non-zero friction, affects the result.*

Consider therefore the following situation: Jirka throws his boule so that its center move along a straight line whose distance from the center of Jarda's boule is $\sqrt{2}R$*, where* $R = 4\,\mathrm{cm}$ *i he radius of the boules. The speed of Jirka's boule immediately before the collision is* $1\,\mathrm{m{\cdot}s^{-1}}$ *The boules are identical, and the collision of the boules is elastic in the normal direction (wit respect to the point of contact of both boules). In the tangential direction, large friction acts so immediately after the collision, the relative speed of the contact points in the tangentia direction is zero. At what angle do the boules roll apart under these assumptions?*

Assume that Jirka's boule is not rotating about its vertical axis just before the collision; an possible rotation in other directions (e.g., due to rolling) will not affect the result. Effectively he angular velocity of Jirka's boule just before the collision can be considered zero.

Jirka wanted to satisfy his craving for hard problems

Problem R.1 ... telescope illumination

4 point

Consider a section (spherical cap) of a concave spherical mirror with radius $R = 200\,\mathrm{cm}$ *an*
adius of base of the cap $r = 150\,\mathrm{mm}$. *In front of this mirror, place a circular* **opaque** *screen c*
adius $\delta = 80\,\mathrm{mm}$ *at a distance* $d = 135\,\mathrm{cm}$ *from the mirror's vertex, such that the center of th*
creen lies on the principal axis of the spherical mirror. Compute the ratio of the illuminated t
he non-illuminated area on the screen's side facing the mirror. Assume that the light is inciden
arallel to the principal axis of the mirror. Feel free to use approximations valid for $r \ll$
vhen solving this problem.

Vlado was thinking about Cassegrain type telescope

Problem R.2 ... shining on a teaspoon

4 point

Consider a section (spherical cap) of a concave spherical mirror with radius $R = 5.5\,\mathrm{cm}$ *an adius of base of the cap* $r = 4.0\,\mathrm{cm}$. *A point source of light, which illuminates the reflectiv urface, is located at a distance* $a = 7.0\,\mathrm{cm}$ *from the vertex of the mirror. A screen is placed a distance* $d = 10\,\mathrm{cm}$ *from the vertex of the mirror and is perpendicular to the principal axis Calculate the radius of the image formed on the screen. Assume that the entire image can fi on the screen.*

Dishes pile up on Vlado, yet he keeps playing with a teaspoon

Problem R.3 ... reflector off the focal point

7 point

A car headlight consists of a parabolic mirror whose shape originated by rotating the parabola z = $v(r/r_0)^2$, $r < r_0$, where $r_0 = 7.0\,\mathrm{cm}$ and $v = 5.0\,\mathrm{cm}$. A negligibly small incoherent ligh source is located at its focus. Assume a screen is placed perpendicularly to the paraboloid axi at a distance $d = 1.0\,\mathrm{m}$ from the top edge of the reflector (plane $z = v$). However, the bulb i not exactly at the focus but on the paraboloid axis at a distance $\mathrm{d}h = -0.1\,\mathrm{cm}$ from it (close to the paraboloid vertex). By how much does the radius of the bright circle at the center o the screen, where both direct light from the source and light reflected by the paraboloid fal increase in this case?

Kuba could not see the road while driving

Problem R.4 ... spotlight at full power

7 point

A headlight in a car consists of a parabolic mirror formed by rotating the parabola $z = v\,(r/r_0)^2$, $r < r_0 = 7.0\,\mathrm{cm}$, where $v = 5.0\,\mathrm{cm}$ about its axis. A negligibly small, incoheren ight source with radiant intensity $I_\Omega = 0.4\,\mathrm{W{\cdot}sr^{-1}}$ in the visible spectrum is located at its focus Assume that at a distance $d = 1.0\,\mathrm{m}$ from the upper edge of the reflector (the plane $z = v$) screen is placed perpendicular to the paraboloid axis. A bright circular region is visible o he screen, illuminated both by rays reflected from the paraboloid and by rays arriving directl rom the source. Determine the magnitude of the jump in intensity at the boundary of thi ircle.

Kuba was thinking about physics while driving